

MyDesign Portfolio

Element L

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

Earlier on (in K3, if you completed it) you provided general lessons learned related to engineering design, and now (in L2) you are asked to provide recommendations for continuing work on this project in particular.

Answer as many of these questions as you can with specific recommendations based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design specific decisions you made). Provide recommendations to someone doing a similar project that might build or continue upon your work. For example: "Taking into account <something we did>, we believe doing _____ would improve _____ because _____."

- What could be done next to go further?
There are many measures that can be taken in order to proceed further. One example is the use of a stronger material such as Polycarbonate glass, also known as Lexan glass. Like Plexiglass, Lexan glass is also transparent but is a much stronger material. Using Lexan glass is excellent due to the fact that it's transparent, which was something that met our Stakeholder's request, and it is much more durable than the Plexiglass.
- What would you keep if you did the project again?
Regardless of the material being used, if we as a team were tasked to start the project from scratch, we would definitely use a transparent material in order to see the cables. We would also use the process of 3d printing our PLA pegs and using the No slip pads to ensure stability. We would also keep the process of using gorilla glue in order to ensure even more stability.
- What would you change?
We as a team would change the shape of our PLA pegs. We would change the pegs into another shape such as a square or a long rectangle shape to

maintain even more stability. If we choose a long rectangle shape, then the material would have to also be transparent order to be able to see the cables

- What would have made this project easier for you to complete?

There are some ways that would've facilitated the project for us. Printing the PLA pegs via onshape caused us the most problems, not because it was difficult but because it was very time consuming. To print 6 pla pegs took about 2 hours and 35 minutes and our class time was about 55 minutes and we needed to print about 60 pegs. If there was a way or mechanism that could have printed out more pegs at a time would've been a lot easier and would've save us a lot of time

In order to earn 5's on Element L, you will need specific and detailed ways the project can be improved and a rationale for why they are needed.

Detailed Plans for Implementation of Improvements (L2)

Now that you have generated a list of specific recommendations with their rationale, go back under each and generate a consistently detailed list about HOW these details will be implemented. Refer back to Element I where you concluded with some of these ideas.

- 1) Use a stronger transparent material such as Lexan (polycarbonate glass) or any other transparent material that has better and more durability than plexiglass
- 2) Either change the shape of our PLA pegs to a square shape or a long rectangle or modify the thickness and shape of the bottom of our pegs
- 3) Make the plexiglass panels longer than 29.5 in x 11.25in to between 35-40 in x 11.25 in
- 4) Replacing the existing materials with stronger and more durable ones in order for the design to last longer
- 5) Improve our design plan and create a more improved blueprint and sketch of our designs in order to have a better idea of how to do complete the assignment